

SCM: Sleep-Consolidated Memory for Language Agents

An End-to-End, Reproducible Memory Lifecycle from Selective Encoding to Contradiction-Safe Consolidation

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Abstract

Large language models are still weak at long-horizon memory management under interference, contradiction, and noise growth. Most deployed memory layers are either context-window extensions or append-only retrieval stores with limited lifecycle control. We present Sleep-Consolidated Memory (SCM), a full memory architecture that combines selective encoding, event-aware association binding, spreading-activation retrieval, staged sleep consolidation (MicroSleep and DeepSleep), adaptive forgetting dynamics, and contradiction-safe version lineage. Unlike isolated module proposals, SCM is implemented and evaluated as one integrated system across six development phases with artifact-backed benchmark suites. On the latest Phase 6 human-memory benchmark, SCM achieves one-shot recall accuracy 1.0000, MicroSleep disambiguation gain 1.0000, DeepSleep disambiguation gain 1.0000, deep-stage noise retention 0.0000, and contradiction-versioning accuracy 1.0000. In a 10-seed stress benchmark with 96 duplicate/noise pairs per seed, lexical retrieval, vector retrieval, and SCM awake-only remain at disambiguation recall 0.0000, while SCM+DeepSleep reaches mean disambiguation recall 0.9052 with mean noise retention 0.0000. In a seven-day long-horizon benchmark, sleep-enabled SCM reaches final family-aware disambiguation recall 0.9812 with final noise retention 0.0000, versus awake-only recall 0.0000 and noise retention 1.0000. A reproducibility pack reruns the full claim stack and reports overall pass. The resulting evidence supports SCM as a rigorous memory-lifecycle baseline for next-generation persistent language agents.

1 Introduction

Persistent memory remains a central bottleneck for language agents. Context-window approaches degrade when relevant information is not near prompt boundaries [4]. Retrieval-augmented systems improve factual access but often behave as growing stores with weak consolidation and weak forgetting

control [3]. System-style memory managers such as MemGPT improve scaling abstractions but are not, by construction, biologically grounded memory lifecycles [6]. Production memory layers such as Mem0 advance personalization but are still primarily awake-time retrieval systems [2].

Human memory is not append-only. It is selective at encoding, bounded in working capacity [5, 1], restructured by sleep [7], and stabilized by active forgetting [9, 8]. This paper operationalizes those principles in a single architecture and evaluates whether sleep-centric lifecycle operations materially improve memory behavior under stress.

The central claim is specific: in controlled duplicate-pressure and long-horizon benchmarks, sleep-enabled SCM produces large disambiguation gains and strong noise suppression that do not appear in awake-only or standard retrieval controls.

2 What This Project Built (A-Z)

SCM was developed as a six-phase program, each phase shipping code, tests, and measurable artifacts.

Table 1: Project build map from Phase 1 to Phase 6.

Phase	Core capability	Evidence path
1	Selective encoding, salience gating, grasp scoring, prediction-error signal	src/core/attention_gate.py ; tests/test_memory_pipeline.py
2	Event compiler and association binder with edge aging and pruning	src/core/event_compiler.py ; src/core/association_binder.py ; research/metrics/phase6_guardrails_latest.json
3	Spreading activation retrieval and hypothesis ranking	src/retrieval/spreading_activation.py ; src/retrieval/hypothesis_ranker.py ; tests/test_phase3_pipeline.py
4	MicroSleep and DeepSleep orchestration, replay, downscaling, synthesis	src/sleep/micro_sleep.py ; src/sleep/deep_sleep.py ; tests/phase4_metrics.py
5	Adaptive forgetting dynamics and contradiction-safe version lineage	src/sleep/forgetting_dynamics.py ; src/core/long_term_memory.py ; tests/test_contradiction_versioning.py
6	Guardrails, human-memory suite, baseline comparison, long horizon, reproducibility pack	research/metrics/phase6_human_memory_latest.json ; research/metrics/baseline_comparison_latest.json ; research/reproducibility/reproducibility_pack_latest.json

This paper describes the integrated system, not a single module in isolation.

3 System Architecture

3.1 Memory Substrates

SCM uses a bounded working memory plus persistent graph long-term memory:

- **Working memory:** capacity-constrained episodic buffer (default capacity 7).
- **Long-term memory:** typed concept graph with weighted relations, persistence, and version lineage metadata.

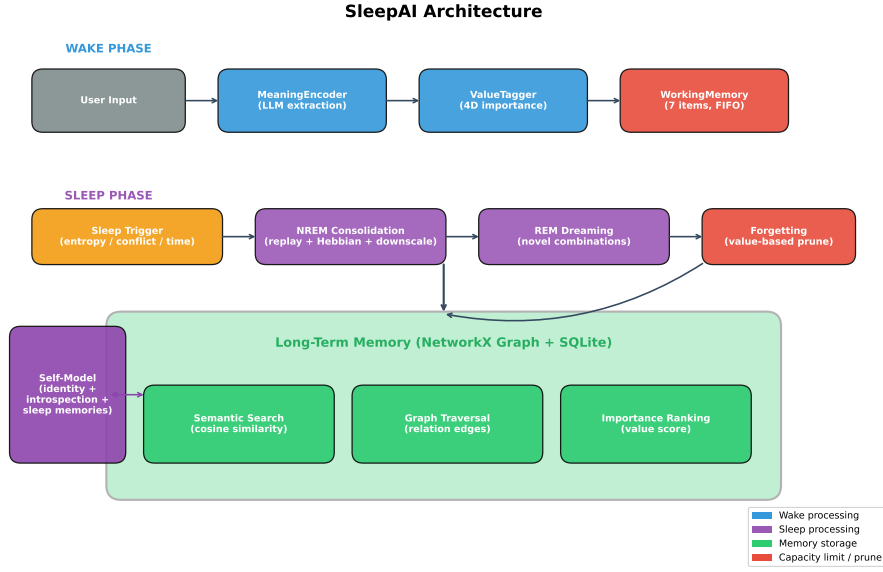


Figure 1: SCM architecture: selective encoding and event binding during wake, staged consolidation and forgetting during sleep, and contradiction-safe version lineage across updates.

3.2 Sleep-State Control

Sleep mode selection is pressure-aware:

- MicroSleep trigger examples: entropy spike, turn interval, conflict pressure.

- DeepSleep trigger examples: high pressure, idle-time window, long session threshold.

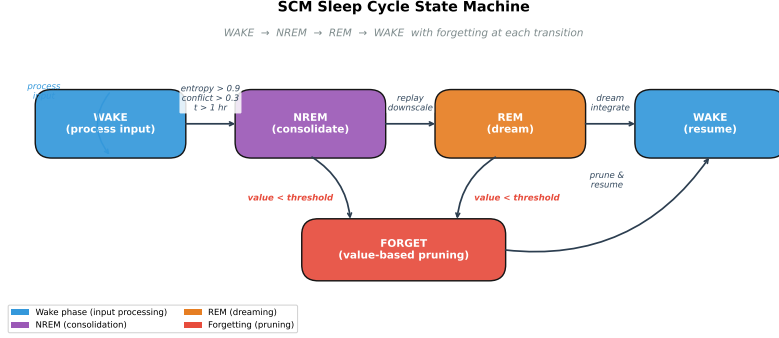


Figure 2: Wake/MicroSleep/DeepSleep control flow used by the orchestrator.

4 Mathematical Formulation of SCM

4.1 Phase 1: Selective Encoding and Grasp

For an input trace x , SCM computes adjusted salience

$$S_{\text{att}}(x) = \text{clip}(w_n N + w_t T + w_e |E| + w_r R + w_p \varepsilon - P_{\text{noise}}, 0, 1), \quad (1)$$

where N, T, E, R, ε are novelty, task relevance, emotional valence, repetition, and prediction error signals. In the current default configuration,

$$(w_n, w_t, w_e, w_r, w_p) = (0.30, 0.35, 0.15, 0.10, 0.10).$$

One-shot grasp is then estimated as

$$G(x) = \text{clip}(\alpha S_{\text{att}} + \beta O_{\text{schema}} + \gamma C_{\text{clarity}} - \delta L_{\text{load}}, 0, 1), \quad (2)$$

with $(\alpha, \beta, \gamma, \delta) = (0.40, 0.30, 0.20, 0.10)$.

4.2 Phase 2: Event-Aware Association Binding

Given concepts c_i, c_j in an event, relation strength increment is

$$\Delta w_{ij} = \eta \cdot \frac{g_i + g_j}{2} \cdot s_{\text{event}} \cdot q_{ij}, \quad (3)$$

with learning rate $\eta = 0.30$ and

$$q_{ij} = \text{clip}\left(0.45 s_{ij}^{\text{sem}} + 0.25 s_{ij}^{\text{tok}} + 0.30 s_{ij}^{\text{ctx}}, 0.05, 1.0\right). \quad (4)$$

Edges are aged by repeated multiplicative decay and pruned when stale and weak.

4.3 Phase 3: Spreading Activation Retrieval

SCM retrieval is cue-seeded graph propagation:

$$A_j^{(t+1)} = \delta A_j^{(t)} + \sum_i A_i^{(t)} W_{ij} \Gamma_{ij}, \quad (5)$$

where $\delta = 0.45$ is spreading decay, W_{ij} relation strength, and Γ_{ij} context gate.

The gate incorporates session/person matches and recency:

$$\Gamma = \text{clip}(k_{\text{session}} \cdot k_{\text{person}} \cdot (0.6 + 0.4r_{\text{recency}}), 0.3, 1.5). \quad (6)$$

Candidate ranking blends cue match and consolidation score, and a recency-boosted variant is used in dedicated retrieval paths.

4.4 Phase 4: Sleep Consolidation

MicroSleep. Concept instability is

$$u(c) = 0.40(1 - r_c) + 0.30e_c + 0.20n_c + 0.10(1 - \tilde{s}_c), \quad (7)$$

where r_c is retention, e_c prediction error, n_c novelty, and $\tilde{s}_c$ normalized strength. Top- k unstable traces are replayed and lightly reinforced; near-duplicate traces are merged.

DeepSleep. NREM performs Hebbian strengthening and downscaling; REM optionally synthesizes new links; forgetting then retires weak traces. A global downscale factor (default 0.90) is applied to concept and relation strengths.

4.5 Phase 5: Forgetting Dynamics and Versioning

Retention score follows

$$R(c) = p_1 G_c + p_2 S_c + p_3 H_c + p_4 D_c + p_5 \text{Rec}_c - p_6 I_c, \quad (8)$$

with default

$$(p_1, p_2, p_3, p_4, p_5, p_6) = (0.22, 0.24, 0.18, 0.18, 0.12, 0.22).$$

Trace-specific decay is

$$\lambda_c = \lambda_0 (1 + 1.5(1 - R(c)) + 0.8I_c + \pi_{\text{version}} + \pi_{\text{state}}), \quad (9)$$

with $\lambda_0 = 0.04$, then

$$s_c \leftarrow \max(0.05, s_c \exp(-\lambda_c \max(0.25, \text{age_hours}/24))). \quad (10)$$

Adaptive thresholding increases forgetting pressure with memory load:

$$\theta_f = \min(0.85, \theta_s + 0.18 \min(1, |M|/100) + 0.05\rho). \quad (11)$$

Contradiction updates use version matching by similarity-plus-context scoring. If score exceeds threshold (default 0.70), the previous concept is archived and linked as a contradiction edge to the new active version.

5 Experimental Design

5.1 Artifact-Centric Protocol

All claims in this paper are tied to current machine-readable artifacts:

- [research/metrics/phase6_human_memory_latest.json](#)
- [research/metrics/baseline_comparison_latest.json](#)
- [research/metrics/long_horizon_latest.json](#)
- [research/metrics/phase6_guardrails_latest.json](#)
- [research/metrics/phase6_backend_smoke_latest.json](#)
- [research/reproducibility/reproducibility_pack_latest.json](#)

5.2 Benchmarks

Human-memory suite. One-shot recall, sleep gain, selective forgetting, contradiction versioning.

Baseline stress comparison. 10 seeds (7001–7010), 96 duplicate/noise pairs per seed. Controls: lexical retrieval, vector retrieval, SCM awake-only baseline, SCM+MicroSleep, SCM+DeepSleep.

Long-horizon benchmark. Seven-day synthetic memory history with repeated interference, extra noise injection, periodic deep sleep, and late contradiction update.

Guardrails and smoke. Phase 2/4/6 status checks, deprecation scan, regression pytest bundle, backend endpoint smoke.

6 Results

6.1 Core Human-Memory Behavior

Table 2: Phase 6 human-memory benchmark (latest artifact).

Metric	Value
One-shot recall accuracy	1.0000
MicroSleep disambiguation gain (absolute)	1.0000
DeepSleep disambiguation gain (absolute)	1.0000
DeepSleep noise retention	0.0000
Contradiction-versioning accuracy	1.0000
Overall status	pass

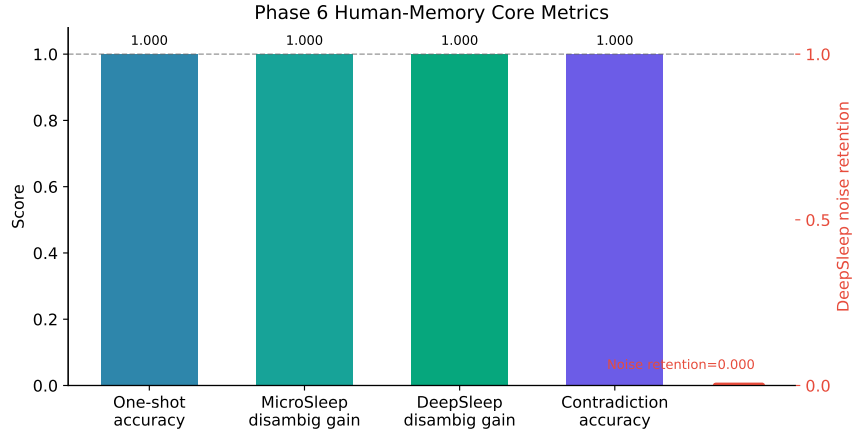


Figure 3: Phase 6 core metrics from the latest human-memory artifact.

6.2 Stress Comparison Against Controls

Table 3: Baseline stress comparison (10 seeds, 96 pairs per seed).

Method	Disambiguation recall	Noise retention	Latency (ms)
Lexical retrieval baseline	0.0000	1.0000	0.5394
Vector retrieval baseline	0.0000	1.0000	15.4428
SCM baseline (no sleep)	0.0000	1.0000	1.9302
SCM + MicroSleep	1.0000	0.0000	1.4698
SCM + DeepSleep	0.9052	0.0000	31.1439

The attribution signal is sharp: in this stress setting, all awake controls remain at 0.0000 disambiguation recall, while both sleep-enabled variants move to high-recall/low-noise regimes.

6.3 Long-Horizon Memory

Table 4: Final-day long-horizon comparison (day 7).

Metric	Awake-only	Sleep-enabled
Family-aware disambiguation recall	0.0000	0.9812
Strict disambiguation recall	0.0000	0.0000
Noise retention	1.0000	0.0000
Anchor hit rate	1.0000	1.0000

Family-aware duplicate recall is the release gate in the current benchmark

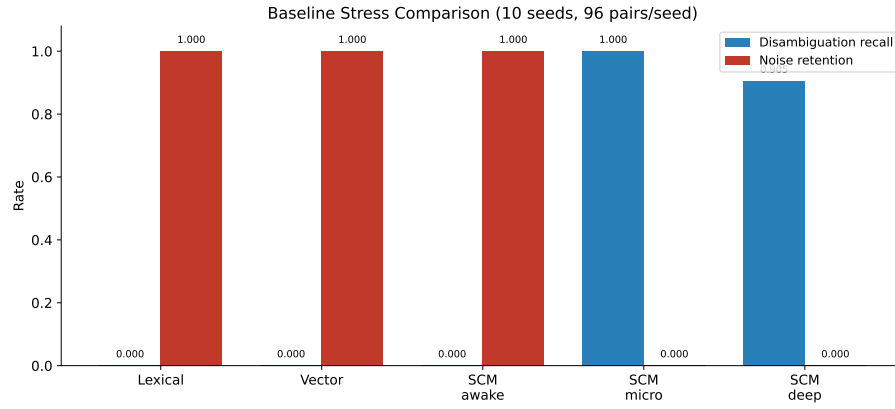


Figure 4: Disambiguation vs. noise-retention behavior across controls and sleep-enabled SCM variants.

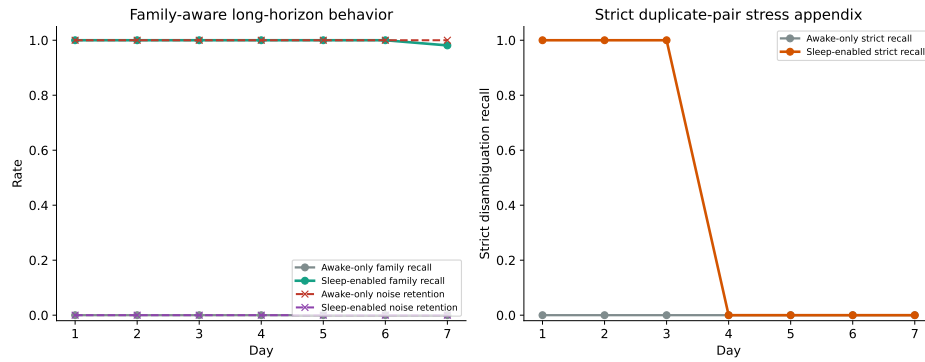


Figure 5: Long-horizon daily curves: family-aware release metric and strict duplicate-pair stress appendix.

design, while strict duplicate-pair recall is preserved as a transparent stress appendix.

6.4 Reliability and Reproducibility

Guardrail status reports pass across Phase 2, Phase 4, and human-memory checks, with zero hits for tracked deprecation patterns. Backend smoke reports 20/20 HTTP success, 20/20 overall pass, mean latency 45.2 ms, p50 22.7 ms, and p90 26.3 ms.

The reproducibility pack reruns baseline comparison, long-horizon benchmark, Phase 6 guardrails, and a smoke pytest bundle in one command chain, reporting overall pass and logging artifact fingerprints.

Table 5: Reproducibility pack execution summary (latest run).

Suite	Result	Duration (ms)
baseline_comparison	pass	58443.53
long_horizon	pass	7296.59
phase6_guardrails	pass	21499.70
smoke_pytests	pass	7291.73

7 Why the Gains Appear

The combined evidence supports a mechanism-level interpretation.

1. **Selective encoding** reduces early contamination from weak traces.
2. **Association binding** creates retrievable structure beyond pure embedding similarity.
3. **Sleep consolidation** reshapes graph topology and strength distributions.
4. **Adaptive forgetting** prunes low-value distractors and preserves strong traces.
5. **Version lineage** prevents contradiction updates from destructive overwrite.

The strongest empirical support for this interpretation is negative evidence from controls: lexical, vector, and awake-only SCM stay at 0.0000 disambiguation recall in the 96-pair stress setting.

8 Limitations and Claim Boundaries

This work is strong on internal reproducibility, but still has clear boundaries.

- Benchmarks are mostly synthetic and harness-driven.
- External head-to-head comparisons remain limited.
- Strict duplicate-pair disambiguation under long horizons remains a known stress weakness.
- Claims are about human-like memory dynamics, not consciousness or sentience.

Accordingly, the strongest defensible claim today is not “human equivalence,” but that a sleep-consolidated lifecycle can outperform awake-only memory pipelines on targeted memory-quality criteria with reproducible evidence.

9 Conclusion

SCM demonstrates that memory quality for language agents improves when memory is treated as a lifecycle rather than as static retrieval storage. Across Phase 6 artifacts, SCM preserves one-shot behavior, executes contradiction-safe updates, suppresses low-value noise, and delivers large disambiguation gains in pressure settings where awake controls fail. The next research milestone is external user validation at scale with broader cross-system baselines.

Reproducibility Statement

Core reproducibility entry points:

- [tests/baseline_comparison.py](#)
- [tests/long_horizon_benchmark.py](#)
- [tests/phase6_guardrails.py](#)
- [tests/reproducibility_pack.py](#)

Latest aggregate evidence:

- [research/reproducibility/reproducibility_pack_latest.json](#)
- [docs/SCM_REPRODUCIBILITY_PACK.md](#)

Algorithm 1 SCM Sleep Orchestration (simplified)

Require: Concepts C , Relations R , Episodes E

```
1: if DeepSleep trigger then
2:    $(C', R') \leftarrow \text{NREM\_consolidate}(C, R, E)$ 
3:    $(C'', R'') \leftarrow \text{REM\_synthesize}(C', R')$ 
4:    $(C^*, R^*) \leftarrow \text{ForgettingDynamics}(C'', R'')$ 
5: else if MicroSleep trigger then
6:    $(C^*, R^*) \leftarrow \text{MicroSleep}(C, R, E)$ 
7: else
8:    $(C^*, R^*) \leftarrow (C, R)$ 
9: end if
10: return  $(C^*, R^*)$ 
```

A High-Level Sleep Orchestration

B Contradiction-Safe Versioning Logic

When a new concept c_{new} is added with versioning enabled, SCM searches for a type-matched candidate c_{old} using similarity plus context bonuses. If the score exceeds threshold:

1. mark c_{old} as archived and no longer current;
2. set $c_{\text{new}}.\text{version_parent} = c_{\text{old}}$ and inherit root lineage;
3. insert a directed contradiction relation $c_{\text{old}} \rightarrow c_{\text{new}}$;
4. keep retrieval on current versions by default, while preserving history for audit.

References

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